Soft Actor-Critic:
Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor

Tuomas Haarnoja [1] **Aurick Zhou** [1] **Pieter Abbeel** [1] **Sergey Levine** [1]

Abstract

Model-free deep reinforcement learning (RL) algorithms have been demonstrated on a range of challenging decision making and control tasks. However, these methods typically suffer from two major challenges: very high sample complexity and brittle convergence properties, which necessitate meticulous hyperparameter tuning. Both of these challenges severely limit the applicability of such methods to complex, real-world domains. In this paper, we propose soft actor-critic, an off-policy actor-critic deep RL algorithm based on the maximum entropy reinforcement learning framework. In this framework, the actor aims to maximize expected reward while also maximizing entropy. That is, to succeed at the task while acting as randomly as possible. Prior deep RL methods based on this framework have been formulated as Q-learning methods. By combining off-policy updates with a stable stochastic actor-critic formulation, our method achieves state-of-the-art performance on a range of continuous control benchmark tasks, outperforming prior on-policy and off-policy methods. Furthermore, we demonstrate that, in contrast to other off-policy algorithms, our approach is very stable, achieving very similar performance across different random seeds.

1. Introduction

Model-free deep reinforcement learning (RL) algorithms have been applied in a range of challenging domains, from games (Mnih et al., 2013; Silver et al., 2016) to robotic control (Schulman et al., 2015). The combination of RL and high-capacity function approximators such as neural networks holds the promise of automating a wide range of decision making and control tasks, but widespread adoption of these methods in real-world domains has been hampered by two major challenges. First, model-free deep RL methods are notoriously expensive in terms of their sample complexity. Even relatively simple tasks can require millions of steps of data collection, and complex behaviors with high-dimensional observations might need substantially more. Second, these methods are often brittle with respect to their hyperparameters: learning rates, exploration constants, and other settings must be set carefully for different problem settings to achieve good results. Both of these challenges severely limit the applicability of model-free deep RL to real-world tasks.

One cause for the poor sample efficiency of deep RL methods is on-policy learning: some of the most commonly used deep RL algorithms, such as TRPO (Schulman et al., 2015), PPO (Schulman et al., 2017b) or A3C (Mnih et al., 2016), require new samples to be collected for each gradient step. This quickly becomes extravagantly expensive, as the number of gradient steps and samples per step needed to learn an effective policy increases with task complexity. Off-policy algorithms aim to reuse past experience. This is not directly feasible with conventional policy gradient formulations, but is relatively straightforward for Q-learning based methods (Mnih et al., 2015). Unfortunately, the combination of off-policy learning and high-dimensional, nonlinear function approximation with neural networks presents a major challenge for stability and convergence (Bhatnagar et al., 2009). This challenge is further exacerbated in continuous state and action spaces, where a separate actor network is often used to perform the maximization in Q-learning. A commonly used algorithm in such settings, deep deterministic policy gradient (DDPG) (Lillicrap et al., 2015), provides for sample-efficient learning but is notoriously challenging to use due to its extreme brittleness and hyperparameter sensitivity (Duan et al., 2016; Henderson et al., 2017).

We explore how to design an efficient and stable model-free deep RL algorithm for continuous state and action spaces. To that end, we draw on the maximum entropy framework, which augments the standard maximum reward

[1]Berkeley Artificial Intelligence Research, University of California, Berkeley, USA. Correspondence to: Tuomas Haarnoja <haarnoja@berkeley.edu>.

Proceedings of the 35th International Conference on Machine Learning, Stockholm, Sweden, PMLR 80, 2018. Copyright 2018 by the author(s).

Soft Actor-Critic：采用随机策略的离策略最大熵深度强化学习

Tuomas Haarnoja [1] Aurick Zhou [1] Pieter Abbeel [1] Sergey Levine [1]

摘要

无模型深度强化学习(RL)算法已在多种具有挑战性的决策与控制任务中得到验证。然而，这类方法通常面临两项主要挑战：样本复杂度很高，以及收敛过程脆弱，因而需要细致地调节超参数。这两项挑战严重限制了此类方法在复杂真实场景中的应用。本文提出软 Actor-Critic(SAC)，这是一种基于最大熵强化学习框架的离策略 Actor-Critic 深度强化学习算法。在这一框架中，Actor 在最大化期望奖励的同时最大化熵，即在完成任务的同时尽可能保持行为的随机性。此前基于这一框架的深度强化学习方法主要采用 Q 学习形式。通过将离策略更新与稳定的随机 Actor-Critic 形式相结合，我们的方法在多种连续控制基准任务上达到了当时的先进水平，优于以往的同策略和离策略方法。此外，我们表明，相较于其他离策略算法，所提方法具有很好的稳定性，在不同随机种子下取得了非常接近的表现。

1. 引言

无模型深度强化学习(RL)算法已应用于多种具有挑战性的领域，从游戏(Mnih et al., 2013; Silver et al., 2016)到机器人控制(Schulman et al., 2015)。将强化学习与神经网络等高容量函数近似器相结合，有望实现多种决策与控制任务的自动化，但这些方法在真实场景中的广泛应用仍受到两项主要挑战的阻碍。首先，无模型深度强化学习的样本复杂度通常很高：即使相对简单的任务，也可能需要采集数百万步数据；对于采用高维观测的复杂行为，所需数据还可能多得多。其次，这类方法通常对超参数较为敏感：为了在不同问题中取得良好结果，需要仔细设置学习率、探索常数等参数。这两项挑战严重限制了无模型深度强化学习在真实任务中的应用。

深度强化学习样本效率较低的一个原因是同策略学习。一些常用算法，如 TRPO(Schulman et al., 2015)、PPO(Schulman et al., 2017b)和 A3C(Mnih et al., 2016)，需要为梯度更新采集新样本。随着任务复杂度增加，学习有效策略所需的梯度更新次数和每次更新的样本数都会增加，代价因而迅速上升。离策略算法旨在复用过去的经验。传统策略梯度形式难以直接做到这一点，而基于 Q 学习的方法相对容易实现(Mnih et al., 2015)。然而，将离策略学习与神经网络提供的高维非线性函数近似结合，会给稳定性和收敛性带来重大挑战(Bhatnagar et al., 2009)。在连续状态与动作空间中，这一挑战更加突出，因为 Q 学习中的最大化操作通常要由独立的 Actor 网络完成。该场景下常用的深度确定性策略梯度(DDPG)(Lillicrap et al., 2015)具有较高样本效率，但其训练过程十分脆弱且对超参数敏感，因此难以使用(Duan et al., 2016; Henderson et al., 2017)。

我们研究如何为连续状态与动作空间设计高效、稳定的无模型深度强化学习算法。为此，我们采用最大熵框架，在标准最大奖励强化学习目标中

[1]Berkeley Artificial Intelligence Research, University of California, Berkeley, USA. Correspondence to: Tuomas Haarnoja <haarnoja@berkeley.edu>.

Proceedings of the 35th International Conference on Machine Learning, Stockholm, Sweden, PMLR 80, 2018. Copyright 2018 by the author(s).

Chinese translation | Haarnoja et al., ICML 2018 | CC BY 4.0 | proceedings.mlr.press/v80/haarnoja18b.html

reinforcement learning objective with an entropy maximization term (Ziebart et al., 2008; Toussaint, 2009; Rawlik et al., 2012; Fox et al., 2016; Haarnoja et al., 2017). Maximum entropy reinforcement learning alters the RL objective, though the original objective can be recovered using a temperature parameter (Haarnoja et al., 2017). More importantly, the maximum entropy formulation provides a substantial improvement in exploration and robustness: as discussed by Ziebart (2010), maximum entropy policies are robust in the face of model and estimation errors, and as demonstrated by (Haarnoja et al., 2017), they improve exploration by acquiring diverse behaviors. Prior work has proposed model-free deep RL algorithms that perform on-policy learning with entropy maximization (O'Donoghue et al., 2016), as well as off-policy methods based on soft Q-learning and its variants (Schulman et al., 2017a; Nachum et al., 2017a; Haarnoja et al., 2017). However, the on-policy variants suffer from poor sample complexity for the reasons discussed above, while the off-policy variants require complex approximate inference procedures in continuous action spaces.

In this paper, we demonstrate that we can devise an off-policy maximum entropy actor-critic algorithm, which we call soft actor-critic (SAC), which provides for both sample-efficient learning and stability. This algorithm extends readily to very complex, high-dimensional tasks, such as the Humanoid benchmark (Duan et al., 2016) with 21 action dimensions, where off-policy methods such as DDPG typically struggle to obtain good results (Gu et al., 2016). SAC also avoids the complexity and potential instability associated with approximate inference in prior off-policy maximum entropy algorithms based on soft Q-learning (Haarnoja et al., 2017). We present a convergence proof for policy iteration in the maximum entropy framework, and then introduce a new algorithm based on an approximation to this procedure that can be practically implemented with deep neural networks, which we call soft actor-critic. We present empirical results that show that soft actor-critic attains a substantial improvement in both performance and sample efficiency over both off-policy and on-policy prior methods. We also compare to twin delayed deep deterministic (TD3) policy gradient algorithm (Fujimoto et al., 2018), which is a concurrent work that proposes a deterministic algorithm that substantially improves on DDPG.

2. Related Work

Our soft actor-critic algorithm incorporates three key ingredients: an actor-critic architecture with separate policy and value function networks, an off-policy formulation that enables reuse of previously collected data for efficiency, and entropy maximization to enable stability and exploration. We review prior works that draw on some of these ideas in this section. Actor-critic algorithms are typically derived starting from policy iteration, which alternates between *policy evaluation*—computing the value function for a policy—and *policy improvement*—using the value function to obtain a better policy (Barto et al., 1983; Sutton & Barto, 1998). In large-scale reinforcement learning problems, it is typically impractical to run either of these steps to convergence, and instead the value function and policy are optimized jointly. In this case, the policy is referred to as the actor, and the value function as the critic. Many actor-critic algorithms build on the standard, on-policy policy gradient formulation to update the actor (Peters & Schaal, 2008), and many of them also consider the entropy of the policy, but instead of maximizing the entropy, they use it as an regularizer (Schulman et al., 2017b; 2015; Mnih et al., 2016; Gruslys et al., 2017). On-policy training tends to improve stability but results in poor sample complexity.

There have been efforts to increase the sample efficiency while retaining robustness by incorporating off-policy samples and by using higher order variance reduction techniques (O'Donoghue et al., 2016; Gu et al., 2016). However, fully off-policy algorithms still attain better efficiency. A particularly popular off-policy actor-critic method, DDPG (Lillicrap et al., 2015), which is a deep variant of the deterministic policy gradient (Silver et al., 2014) algorithm, uses a Q-function estimator to enable off-policy learning, and a deterministic actor that maximizes this Q-function. As such, this method can be viewed both as a deterministic actor-critic algorithm and an approximate Q-learning algorithm. Unfortunately, the interplay between the deterministic actor network and the Q-function typically makes DDPG extremely difficult to stabilize and brittle to hyperparameter settings (Duan et al., 2016; Henderson et al., 2017). As a consequence, it is difficult to extend DDPG to complex, high-dimensional tasks, and on-policy policy gradient methods still tend to produce the best results in such settings (Gu et al., 2016). Our method instead combines off-policy actor-critic training with a *stochastic* actor, and further aims to maximize the entropy of this actor with an entropy maximization objective. We find that this actually results in a considerably more stable and scalable algorithm that, in practice, exceeds both the efficiency and final performance of DDPG. A similar method can be derived as a zero-step special case of stochastic value gradients (SVG(0)) (Heess et al., 2015). However, SVG(0) differs from our method in that it optimizes the standard maximum expected return objective, and it does not make use of a separate value network, which we found to make training more stable.

Maximum entropy reinforcement learning optimizes policies to maximize both the expected return and the expected entropy of the policy. This framework has been used in many contexts, from inverse reinforcement learning (Ziebart et al., 2008) to optimal control (Todorov, 2008; Toussaint, 2009; Rawlik et al., 2012). In guided policy

加入熵最大化项(Ziebart et al., 2008; Toussaint, 2009; Rawlik et al., 2012; Fox et al., 2016; Haarnoja et al., 2017)。最大熵强化学习改变了强化学习的目标，但仍可通过温度参数恢复原目标(Haarnoja et al., 2017)。更重要的是，最大熵形式能够显著改善探索和鲁棒性：Ziebart(2010)指出，最大熵策略面对模型误差和估计误差具有鲁棒性；相关研究表明，它能通过获得多样化行为来改善探索 (Haarnoja et al., 2017)。已有工作提出了结合熵最大化的同策略无模型深度强化学习算法(O’Donoghue et al., 2016)，以及基于软 Q 学习及其变体的离策略方法(Schulman et al., 2017a; Nachum et al., 2017a; Haarnoja et al., 2017)。不过，同策略变体仍受到上述样本复杂度问题的限制，而离策略变体在连续动作空间中需要复杂的近似推断过程。

本文表明，可以设计一种兼顾样本效率与稳定性的离策略最大熵 Actor-Critic 算法，我们将其称为软 Actor-Critic(SAC)。该算法能够直接扩展到复杂的高维任务，例如具有 21 维动作的 Humanoid 基准(Duan et al., 2016)；而 DDPG 等离策略方法通常难以在这类任务上取得良好结果(Gu et al., 2016)。SAC 还避免了以往基于软 Q 学习的离策略最大熵方法中，近似推断所带来的复杂性和潜在不稳定性(Haarnoja et al., 2017)。我们先证明最大熵框架下策略迭代的收敛性，再通过近似这一过程，提出可以采用深度神经网络实际实现的 SAC 算法。实验表明，相较于已有同策略和离策略方法，SAC 的性能与样本效率均有显著提升。我们还与同期提出的双延迟深度确定性策略梯度(TD3)算法(Fujimoto et al., 2018)进行了比较，该确定性算法对 DDPG 作出了显著改进。

2. 相关工作

我们的 SAC 算法包含三个关键组成部分：策略网络和价值函数网络相互独立的 Actor-Critic 架构；通过复用已采集数据来提高效率的离策略形式；以及用于促进稳定性和探索的熵最大化。本节回顾采用这些思路的相关工作。Actor-Critic 算法通常基于策略迭代推导而来。策略迭代交替执行策略评估，即计算策略的价值函数，以及策略改进，即利用价值函数获得更好的策略(Barto et al., 1983; Sutton & Barto, 1998)。在大规模强化学习问题中，将任一步骤执行至收敛通常并不现实，因此实际中会联合优化价值函数与策略。此时，策略称为 Actor，价值函数称为 Critic。许多 Actor-Critic 算法采用标准同策略策略梯度形式更新 Actor(Peters & Schaal, 2008)；其中不少方法也考虑策略熵，但将其作为正则项，而非直接将熵最大化(Schulman et al., 2017b; 2015; Mnih et al., 2016; Gruslys et al., 2017)。同策略训练通常能提高稳定性，但样本复杂度较高。

已有研究尝试引入离策略样本和高阶方差缩减技术，在保持鲁棒性的同时提高样本效率(O’Donoghue et al., 2016; Gu et al., 2016)。不过，完全离策略的算法仍具有更高效率。DDPG(Lillicrap et al., 2015)是一种广泛使用的离策略 Actor-Critic 方法，也是确定性策略梯度算法(Silver et al., 2014)的深度版本。它利用 Q 函数估计器实现离策略学习，并由确定性 Actor 最大化该 Q 函数。因此，它既可视为确定性 Actor-Critic 算法，也可视为近似 Q 学习算法。然而，确定性 Actor 网络与 Q 函数之间的相互作用，通常使 DDPG 难以稳定，并对超参数设置敏感(Duan et al., 2016; Henderson et al., 2017)。因此，DDPG 很难扩展到复杂高维任务，而同策略策略梯度方法在这些场景下通常仍能获得最佳结果(Gu et al., 2016)。我们的方法将离策略 Actor-Critic 训练与随机 Actor 相结合，并通过熵最大化目标进一步最大化该 Actor 的熵。我们发现，这能得到明显更加稳定且易于扩展的算法，在实践中的效率和最终性能均超过 DDPG。随机价值梯度的零步特例 SVG(0)(Heess et al., 2015)也可推导出类似方法，但 SVG(0) 优化的是标准最大期望回报目标，而且没有独立价值网络；我们发现，独立价值网络能够使训练更稳定。

最大熵强化学习通过优化策略，同时最大化期望回报和策略的期望熵。这一框架已用于多种场景，从逆强化学习(Ziebart et al., 2008)到最优控制(Todorov, 2008; Toussaint, 2009; Rawlik et al., 2012)。在引导策略搜索

search (Levine & Koltun, 2013; Levine et al., 2016), the maximum entropy distribution is used to guide policy learning towards high-reward regions. More recently, several papers have noted the connection between Q-learning and policy gradient methods in the framework of maximum entropy learning (O'Donoghue et al., 2016; Haarnoja et al., 2017; Nachum et al., 2017a; Schulman et al., 2017a). While most of the prior model-free works assume a discrete action space, Nachum et al. (2017b) approximate the maximum entropy distribution with a Gaussian and Haarnoja et al. (2017) with a sampling network trained to draw samples from the optimal policy. Although the soft Q-learning algorithm proposed by Haarnoja et al. (2017) has a value function and actor network, it is not a true actor-critic algorithm: the Q-function is estimating the optimal Q-function, and the actor does not directly affect the Q-function except through the data distribution. Hence, Haarnoja et al. (2017) motivates the actor network as an approximate sampler, rather than the actor in an actor-critic algorithm. Crucially, the convergence of this method hinges on how well this sampler approximates the true posterior. In contrast, we prove that our method converges to the optimal policy from a given policy class, regardless of the policy parameterization. Furthermore, these prior maximum entropy methods generally do not exceed the performance of state-of-the-art off-policy algorithms, such as DDPG, when learning from scratch, though they may have other benefits, such as improved exploration and ease of fine-tuning. In our experiments, we demonstrate that our soft actor-critic algorithm does in fact exceed the performance of prior state-of-the-art off-policy deep RL methods by a wide margin.

3. Preliminaries

We first introduce notation and summarize the standard and maximum entropy reinforcement learning frameworks.

3.1. Notation

We address policy learning in continuous action spaces. We consider an infinite-horizon Markov decision process (MDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, p, r)$, where the state space $\mathcal{S}$ and the action space $\mathcal{A}$ are continuous, and the unknown state transition probability $p : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, \infty)$ represents the probability density of the next state $\mathbf{s}_{t+1} \in \mathcal{S}$ given the current state $\mathbf{s}_t \in \mathcal{S}$ and action $\mathbf{a}_t \in \mathcal{A}$. The environment emits a bounded reward $r : \mathcal{S} \times \mathcal{A} \rightarrow [r_{\min}, r_{\max}]$ on each transition. We will use $\rho_\pi(\mathbf{s}_t)$ and $\rho_\pi(\mathbf{s}_t, \mathbf{a}_t)$ to denote the state and state-action marginals of the trajectory distribution induced by a policy $\pi(\mathbf{a}_t|\mathbf{s}_t)$.

3.2. Maximum Entropy Reinforcement Learning

Standard RL maximizes the expected sum of rewards $\sum_t \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t)]$. We will consider a more general maximum entropy objective (see e.g. Ziebart (2010)), which favors stochastic policies by augmenting the objective with the expected entropy of the policy over $\rho_\pi(\mathbf{s}_t)$:

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} \left[r(\mathbf{s}_t, \mathbf{a}_t) + \alpha \mathcal{H}(\pi(\cdot\,|\mathbf{s}_t)) \right]. \quad (1)$$

The temperature parameter α determines the relative importance of the entropy term against the reward, and thus controls the stochasticity of the optimal policy. The maximum entropy objective differs from the standard maximum expected reward objective used in conventional reinforcement learning, though the conventional objective can be recovered in the limit as $\alpha \rightarrow 0$. For the rest of this paper, we will omit writing the temperature explicitly, as it can always be subsumed into the reward by scaling it by α^{-1}.

This objective has a number of conceptual and practical advantages. First, the policy is incentivized to explore more widely, while giving up on clearly unpromising avenues. Second, the policy can capture multiple modes of near-optimal behavior. In problem settings where multiple actions seem equally attractive, the policy will commit equal probability mass to those actions. Lastly, prior work has observed improved exploration with this objective (Haarnoja et al., 2017; Schulman et al., 2017a), and in our experiments, we observe that it considerably improves learning speed over state-of-art methods that optimize the conventional RL objective function. We can extend the objective to infinite horizon problems by introducing a discount factor γ to ensure that the sum of expected rewards and entropies is finite. Writing down the maximum entropy objective for the infinite horizon discounted case is more involved (Thomas, 2014) and is deferred to Appendix A.

Prior methods have proposed directly solving for the optimal Q-function, from which the optimal policy can be recovered (Ziebart et al., 2008; Fox et al., 2016; Haarnoja et al., 2017). We will discuss how we can devise a soft actor-critic algorithm through a policy iteration formulation, where we instead evaluate the Q-function of the current policy and update the policy through an *off-policy* gradient update. Though such algorithms have previously been proposed for conventional reinforcement learning, our method is, to our knowledge, the first off-policy actor-critic method in the maximum entropy reinforcement learning framework.

4. From Soft Policy Iteration to Soft Actor-Critic

Our off-policy soft actor-critic algorithm can be derived starting from a maximum entropy variant of the policy iteration method. We will first present this derivation, verify that the corresponding algorithm converges to the optimal policy from its density class, and then present a practical

(Levine & Koltun, 2013; Levine et al., 2016)中，最大熵分布用于引导策略学习趋向高奖励区域。近期，多篇论文指出了最大熵学习框架中 Q 学习与策略梯度方法之间的联系(O’ Donoghue et al., 2016; Haarnoja et al., 2017; Nachum et al., 2017a; Schulman et al., 2017a)。以往多数无模型工作假设动作空间离散；Nachum et al. (2017b) 用高斯分布近似最大熵分布，Haarnoja et al. (2017) 则训练采样网络，从最优策略中抽取样本。虽然 Haarnoja et al. (2017) 提出的软 Q 学习具有价值函数和 Actor 网络，它并非真正的 Actor-Critic 算法：Q 函数估计的是最优 Q 函数，Actor 除了通过数据分布之外，并不直接影响 Q 函数。因此，Haarnoja et al. (2017) 将 Actor 网络解释为近似采样器，而非 Actor-Critic 算法中的 Actor。关键在于，该方法能否收敛取决于采样器对真实后验的近似程度。相比之下，我们证明，无论采用何种策略参数化，所提方法都能收敛到给定策略类中的最优策略。此外，这些已有最大熵方法在从零学习时，通常无法超过 DDPG 等先进离策略算法，尽管它们可能改善探索并便于微调。实验表明，我们的 SAC 算法在性能上大幅超过了以往先进的离策略深度强化学习方法。

3. 预备知识

我们首先介绍记号，并概述标准强化学习和最大熵强化学习框架。

3.1. 记号

本文研究连续动作空间中的策略学习。考虑由元组$(\mathcal{S}, \mathcal{A}, p, r)$定义的无限时域马尔可夫决策过程(MDP)，其中状态空间 $\mathcal{S}$ 和动作空间 $\mathcal{A}$ 均为连续空间。未知的状态转移概率 $p : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, \infty)$ 表示在给定当前状态 $\mathbf{s}_t \in \mathcal{S}$ 和动作 $\mathbf{a}_t \in \mathcal{A}$ 时，下一状态 $\mathbf{s}_{t+1} \in \mathcal{S}$ 的概率密度。环境在每次转移时产生有界奖励 $r : \mathcal{S} \times \mathcal{A} \rightarrow [r_{\min}, r_{\max}]$。我们用 $\rho_\pi(\mathbf{s}_t)$ 和 $\rho_\pi(\mathbf{s}_t, \mathbf{a}_t)$ 分别表示策略 $\pi(\mathbf{a}_t|\mathbf{s}_t)$ 所诱导轨迹分布的状态边缘分布和状态—动作边缘分布。

3.2. 最大熵强化学习

标准强化学习最大化期望累计奖励 $\sum_t \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t)]$。我们考虑更一般的最大熵目标(例如 Ziebart (2010))：在目标中加入关于 $\rho_\pi(\mathbf{s}_t)$ 的策略期望熵，从而鼓励随机策略：

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} \left[r(\mathbf{s}_t, \mathbf{a}_t) + \alpha \mathcal{H}(\pi(\cdot\,|\mathbf{s}_t)) \right]. \quad (1)$$

温度参数 α 决定熵项相对于奖励项的重要程度，从而控制最优策略的随机性。最大熵目标与传统强化学习使用的最大期望奖励目标有所不同，但当 $\alpha \rightarrow 0$ 时，可以在极限意义下恢复传统目标。下文不再显式写出温度参数，因为通过将奖励乘以 α^{-1}，总能把温度吸收到奖励中。

这一目标具有多项概念和实践优势。首先，它鼓励策略更广泛地探索，同时放弃明显缺乏前景的方向。其次，策略能够捕捉近最优行为的多个模态。当多种动作看起来同样有吸引力时，策略会为这些动作分配相同的概率质量。最后，已有研究观察到该目标能够改善探索(Haarnoja et al., 2017; Schulman et al., 2017a)；我们的实验也表明，相较于优化传统强化学习目标的先进方法，它能显著提高学习速度。引入折扣因子 γ，使期望奖励与熵之和保持有限，即可将该目标扩展到无限时域问题。无限时域折扣情形下的最大熵目标表达更为复杂(Thomas, 2014)，详见附录 A。

已有方法提出直接求解最优 Q 函数，并据此恢复最优策略(Ziebart et al., 2008; Fox et al., 2016; Haarnoja et al., 2017)。我们将讨论如何通过策略迭代构造 SAC 算法：评估当前策略的 Q 函数，再通过离策略梯度更新改进策略。尽管传统强化学习中已有此类算法，但据我们所知，本文方法是最大熵强化学习框架下首个离策略 Actor-Critic 方法。

4. 从软策略迭代到 Soft Actor-Critic

我们的离策略 SAC 算法可由策略迭代的最大熵变体推导而来。我们首先给出推导，验证该算法能够收敛到所属概率密度类中的最优策略，随后给出

deep reinforcement learning algorithm based on this theory.

4.1. Derivation of Soft Policy Iteration

We will begin by deriving soft policy iteration, a general algorithm for learning optimal maximum entropy policies that alternates between policy evaluation and policy improvement in the maximum entropy framework. Our derivation is based on a tabular setting, to enable theoretical analysis and convergence guarantees, and we extend this method into the general continuous setting in the next section. We will show that soft policy iteration converges to the optimal policy within a set of policies which might correspond, for instance, to a set of parameterized densities.

In the policy evaluation step of soft policy iteration, we wish to compute the value of a policy π according to the maximum entropy objective in Equation 1. For a fixed policy, the soft Q-value can be computed iteratively, starting from any function $Q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and repeatedly applying a modified Bellman backup operator $\mathcal{T}^\pi$ given by

$$\mathcal{T}^\pi Q(\mathbf{s}_t, \mathbf{a}_t) \triangleq r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p}\left[V(\mathbf{s}_{t+1})\right], \quad (2)$$

where

$$V(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t \sim \pi}\left[Q(\mathbf{s}_t, \mathbf{a}_t) - \log \pi(\mathbf{a}_t|\mathbf{s}_t)\right] \quad (3)$$

is the soft state value function. We can obtain the soft value function for any policy π by repeatedly applying $\mathcal{T}^\pi$ as formalized below.

Lemma 1 (Soft Policy Evaluation)**.** *Consider the soft Bellman backup operator $\mathcal{T}^\pi$ in Equation 2 and a mapping $Q^0 : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ with $|\mathcal{A}| < \infty$, and define $Q^{k+1} = \mathcal{T}^\pi Q^k$. Then the sequence Q^k will converge to the soft Q-value of π as $k \rightarrow \infty$.*

Proof. See Appendix B.1. □

In the policy improvement step, we update the policy towards the exponential of the new Q-function. This particular choice of update can be guaranteed to result in an improved policy in terms of its soft value. Since in practice we prefer policies that are tractable, we will additionally restrict the policy to some set of policies Π, which can correspond, for example, to a parameterized family of distributions such as Gaussians. To account for the constraint that $\pi \in \Pi$, we project the improved policy into the desired set of policies. While in principle we could choose any projection, it will turn out to be convenient to use the information projection defined in terms of the Kullback-Leibler divergence. In the other words, in the policy improvement step, for each state, we update the policy according to

$$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} \mathrm{D}_{\mathrm{KL}}\left(\pi'(\cdot\,|\mathbf{s}_t) \,\middle\|\, \frac{\exp\left(Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right)}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)}\right). \quad (4)$$

The partition function $Z^{\pi_{\text{old}}}(\mathbf{s}_t)$ normalizes the distribution, and while it is intractable in general, it does not contribute to the gradient with respect to the new policy and can thus be ignored, as noted in the next section. For this projection, we can show that the new, projected policy has a higher value than the old policy with respect to the objective in Equation 1. We formalize this result in Lemma 2.

Lemma 2 (Soft Policy Improvement)**.** *Let $\pi_{\text{old}} \in \Pi$ and let π_{new} be the optimizer of the minimization problem defined in Equation 4. Then $Q^{\pi_{\text{new}}}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi_{\text{old}}}(\mathbf{s}_t, \mathbf{a}_t)$ for all $(\mathbf{s}_t, \mathbf{a}_t) \in \mathcal{S} \times \mathcal{A}$ with $|\mathcal{A}| < \infty$.*

Proof. See Appendix B.2. □

The full soft policy iteration algorithm alternates between the soft policy evaluation and the soft policy improvement steps, and it will provably converge to the optimal maximum entropy policy among the policies in Π (Theorem 1). Although this algorithm will provably find the optimal solution, we can perform it in its exact form only in the tabular case. Therefore, we will next approximate the algorithm for continuous domains, where we need to rely on a function approximator to represent the Q-values, and running the two steps until convergence would be computationally too expensive. The approximation gives rise to a new practical algorithm, called soft actor-critic.

Theorem 1 (Soft Policy Iteration)**.** *Repeated application of soft policy evaluation and soft policy improvement from any $\pi \in \Pi$ converges to a policy π^* such that $Q^{\pi^*}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$ for all $\pi \in \Pi$ and $(\mathbf{s}_t, \mathbf{a}_t) \in \mathcal{S} \times \mathcal{A}$, assuming $|\mathcal{A}| < \infty$.*

Proof. See Appendix B.3. □

4.2. Soft Actor-Critic

As discussed above, large continuous domains require us to derive a practical approximation to soft policy iteration. To that end, we will use function approximators for both the Q-function and the policy, and instead of running evaluation and improvement to convergence, alternate between optimizing both networks with stochastic gradient descent. We will consider a parameterized state value function $V_\psi(\mathbf{s}_t)$, soft Q-function $Q_\theta(\mathbf{s}_t, \mathbf{a}_t)$, and a tractable policy $\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$. The parameters of these networks are ψ, θ, and ϕ. For example, the value functions can be modeled as expressive neural networks, and the policy as a Gaussian with mean and covariance given by neural networks. We will next derive update rules for these parameter vectors.

The state value function approximates the soft value. There is no need in principle to include a separate function approximator for the state value, since it is related to the Q-function and policy according to Equation 3. This quantity can be

基于这一理论的实用深度强化学习算法。

4.1. 软策略迭代的推导

我们先推导软策略迭代。这是一种学习最优最大熵策略的通用算法，在最大熵框架中交替执行策略评估与策略改进。为便于理论分析并给出收敛保证，推导从表格型情形出发；下一节再扩展到一般连续情形。我们将证明，软策略迭代会收敛到给定策略集合中的最优策略，该集合例如可以是一组参数化概率密度。

在软策略迭代的策略评估步骤中，我们希望依据式 1 的最大熵目标计算策略 π 的价值。对于固定策略，可以从任意函数 $Q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ 出发，反复应用经过修改的贝尔曼备份算子 $\mathcal{T}^{\pi}$，迭代计算软 Q 值。该算子定义为

$$\mathcal{T}^{\pi} Q(\mathbf{s}_t, \mathbf{a}_t) \triangleq r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p}\left[V(\mathbf{s}_{t+1})\right], \quad (2)$$

其中，

$$V(\mathbf{s}_t) = \mathbb{E}_{\mathbf{a}_t \sim \pi}\left[Q(\mathbf{s}_t, \mathbf{a}_t) - \log \pi(\mathbf{a}_t | \mathbf{s}_t)\right] \quad (3)$$

为软状态价值函数。反复应用 $\mathcal{T}^{\pi}$，即可得到任意策略 π 的软价值函数，其形式化结论如下。

引理 1(软策略评估)。考虑式 2 中的软贝尔曼备份算子 $\mathcal{T}^{\pi}$，以及满足 $|\mathcal{A}| < \infty$ 的映射 $Q^0 : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$，定义 $Q^{k+1} = \mathcal{T}^{\pi} Q^k$。则当 $k \rightarrow \infty$ 时，序列 Q^k 收敛到策略 π 的软 Q 值。

证明。见附录 B.1。 □

在策略改进步骤中，我们将策略向新 Q 函数的指数形式更新。可以保证，这种更新会得到软价值更高的策略。由于实践中倾向于采用便于计算的策略，我们进一步将策略限制在集合 Π 中，例如高斯分布等参数化分布族。为满足约束 $\pi \in \Pi$，需要把改进后的策略投影到目标策略集合中。原则上可以选择任意投影，但采用基于 Kullback-Leibler 散度定义的信息投影会更方便。换言之，在策略改进步骤中，对每个状态按下式更新策略:

$$\pi_{\text{new}} = \arg\min_{\pi' \in \Pi} \mathrm{D}_{\mathrm{KL}}\left(\pi'(\cdot | \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot)\right)}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)}\right). \quad (4)$$

配分函数 $Z^{\pi_{\text{old}}}(\mathbf{s}_t)$ 用于归一化该分布。虽然它通常难以计算，但不会影响关于新策略的梯度，因此可以忽略，下一节将进一步说明。对于这一投影，我们可以证明，就式 1 的目标而言，投影后的新策略比旧策略具有更高价值。引理 2 给出形式化结论。

引理 2(软策略改进)。设 $\pi_{\text{old}} \in \Pi$，并令 π_{new} 为式 4 所定义最小化问题的最优解。当 $|\mathcal{A}| < \infty$ 时，对所有($\mathbf{s}_t, \mathbf{a}_t$)$\in \mathcal{S} \times \mathcal{A}$，都有 $Q^{\pi_{\text{new}}}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi_{\text{old}}}(\mathbf{s}_t, \mathbf{a}_t)$。

证明。见附录 B.2。 □

完整的软策略迭代算法交替执行软策略评估与软策略改进。可以证明，它将收敛到策略集合 Π (定理 1) 中的最优最大熵策略。虽然该算法能够找到最优解，但只有在表格型情形下才能精确执行。因此，接下来将其近似用于连续域：此时需要用函数近似器表示 Q 值，而把两个步骤分别执行至收敛的计算成本过高。这一近似产生了新的实用算法，即软 Actor-Critic。

定理 1(软策略迭代)。从任意 $\pi \in \Pi$ 出发，反复执行软策略评估与软策略改进，将收敛到策略 π^*。假设 $|\mathcal{A}| < \infty$，则对所有 $\pi \in \Pi$ 以及($\mathbf{s}_t, \mathbf{a}_t$)$\in \mathcal{S} \times \mathcal{A}$，都有 $Q^{\pi^*}(\mathbf{s}_t, \mathbf{a}_t) \geq Q^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$。

证明。见附录 B.3。 □

4.2. Soft Actor-Critic

如上所述，大规模连续域需要采用软策略迭代的实用近似。为此，我们对 Q 函数和策略都使用函数近似器，并通过随机梯度下降交替优化两个网络，而非将评估与改进分别执行至收敛。考虑参数化状态价值函数 $V_{\psi}(\mathbf{s}_t)$、软 Q 函数 $Q_{\theta}(\mathbf{s}_t, \mathbf{a}_t)$ 和便于计算的策略 $\pi_{\phi}(\mathbf{a}_t | \mathbf{s}_t)$。这些网络的参数为 ψ, θ 和 ϕ。例如，价值函数可用表达能力较强的神经网络建模，策略则可采用均值与协方差由神经网络给出的高斯分布。下面推导这些参数向量的更新规则。

状态价值函数用于近似软价值。原则上，无需为状态价值单独设置函数近似器，因为根据式 3，它与 Q 函数及策略相联系。这个量可以

estimated from a single action sample from the current policy without introducing a bias, but in practice, including a separate function approximator for the soft value can stabilize training and is convenient to train simultaneously with the other networks. The soft value function is trained to minimize the squared residual error

$$J_V(\psi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\frac{1}{2} \left(V_\psi(\mathbf{s}_t) - \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} \left[Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) \right] \right)^2 \right] \tag{5}$$

where $\mathcal{D}$ is the distribution of previously sampled states and actions, or a replay buffer. The gradient of Equation 5 can be estimated with an unbiased estimator

$$\hat{\nabla}_\psi J_V(\psi) = \nabla_\psi V_\psi(\mathbf{s}_t) \left(V_\psi(\mathbf{s}_t) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t) + \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) \right), \tag{6}$$

where the actions are sampled according to the current policy, instead of the replay buffer. The soft Q-function parameters can be trained to minimize the soft Bellman residual

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}} \left[\frac{1}{2} \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \hat{Q}(\mathbf{s}_t, \mathbf{a}_t) \right)^2 \right], \tag{7}$$

with

$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p} \left[V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right], \tag{8}$$

which again can be optimized with stochastic gradients

$$\hat{\nabla}_\theta J_Q(\theta) = \nabla_\theta Q_\theta(\mathbf{a}_t, \mathbf{s}_t) \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - r(\mathbf{s}_t, \mathbf{a}_t) - \gamma V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right). \tag{9}$$

The update makes use of a target value network $V_{\bar{\psi}}$, where $\bar{\psi}$ can be an exponentially moving average of the value network weights, which has been shown to stabilize training (Mnih et al., 2015). Alternatively, we can update the target weights to match the current value function weights periodically (see Appendix E). Finally, the policy parameters can be learned by directly minimizing the expected KL-divergence in Equation 4:

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\mathrm{D}_{\mathrm{KL}} \left(\pi_\phi(\cdot | \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(Q_\theta(\mathbf{s}_t, \cdot) \right)}{Z_\theta(\mathbf{s}_t)} \right) \right]. \tag{10}$$

There are several options for minimizing J_π. A typical solution for policy gradient methods is to use the likelihood ratio gradient estimator (Williams, 1992), which does not require backpropagating the gradient through the policy and the target density networks. However, in our case, the target density is the Q-function, which is represented by a neural network an can be differentiated, and it is thus convenient to apply the reparameterization trick instead, resulting in a lower variance estimator. To that end, we reparameterize the policy using a neural network transformation

$$\mathbf{a}_t = f_\phi(\epsilon_t; \mathbf{s}_t), \tag{11}$$

Algorithm 1 Soft Actor-Critic

Initialize parameter vectors ψ, $\bar{\psi}$, θ, ϕ.
for each iteration **do**
  for each environment step **do**
    $\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t | \mathbf{s}_t)$
    $\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_V \hat{\nabla}_\psi J_V(\psi)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$
    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$
    $\bar{\psi} \leftarrow \tau \psi + (1 - \tau) \bar{\psi}$
  end for
end for

where ϵ_t is an input noise vector, sampled from some fixed distribution, such as a spherical Gaussian. We can now rewrite the objective in Equation 10 as

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}, \epsilon_t \sim \mathcal{N}} \left[\log \pi_\phi(f_\phi(\epsilon_t; \mathbf{s}_t) | \mathbf{s}_t) - Q_\theta(\mathbf{s}_t, f_\phi(\epsilon_t; \mathbf{s}_t)) \right], \tag{12}$$

where π_ϕ is defined implicitly in terms of f_ϕ, and we have noted that the partition function is independent of ϕ and can thus be omitted. We can approximate the gradient of Equation 12 with

$$\begin{aligned} \hat{\nabla}_\phi J_\pi(\phi) = {} & \nabla_\phi \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) \\ & + \left(\nabla_{\mathbf{a}_t} \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) - \nabla_{\mathbf{a}_t} Q(\mathbf{s}_t, \mathbf{a}_t) \right) \nabla_\phi f_\phi(\epsilon_t; \mathbf{s}_t), \end{aligned} \tag{13}$$

where $\mathbf{a}_t$ is evaluated at $f_\phi(\epsilon_t; \mathbf{s}_t)$. This unbiased gradient estimator extends the DDPG style policy gradients (Lillicrap et al., 2015) to any tractable stochastic policy.

Our algorithm also makes use of two Q-functions to mitigate positive bias in the policy improvement step that is known to degrade performance of value based methods (Hasselt, 2010; Fujimoto et al., 2018). In particular, we parameterize two Q-functions, with parameters θ_i, and train them independently to optimize $J_Q(\theta_i)$. We then use the minimum of the the Q-functions for the value gradient in Equation 6 and policy gradient in Equation 13, as proposed by Fujimoto et al. (2018). Although our algorithm can learn challenging tasks, including a 21-dimensional Humanoid, using just a single Q-function, we found two Q-functions significantly speed up training, especially on harder tasks. The complete algorithm is described in Algorithm 1. The method alternates between collecting experience from the environment with the current policy and updating the function approximators using the stochastic gradients from batches sampled from a replay buffer. In practice, we take a single environment step followed by one or several gradient steps (see Appendix D

通过从当前策略抽取一个动作样本来无偏估计，但在实践中，为软价值单独设置函数近似器能够稳定训练，也便于与其他网络同时训练。训练软价值函数时，最小化下列平方残差：

$$J_V(\psi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\frac{1}{2} \left(V_\psi(\mathbf{s}_t) - \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} \left[Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) \right] \right)^2 \right] \tag{5}$$

其中，$\mathcal{D}$ 是此前采样的状态与动作的分布，也可视为经验回放缓冲区。式 5 的梯度可用下列无偏估计器估计：

$$\hat{\nabla}_\psi J_V(\psi) = \nabla_\psi V_\psi(\mathbf{s}_t) \left(V_\psi(\mathbf{s}_t) - Q_\theta(\mathbf{s}_t, \mathbf{a}_t) + \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) \right), \tag{6}$$

其中，动作从当前策略采样，而非从经验回放缓冲区采样。软 Q 函数的参数通过最小化软贝尔曼残差来训练：

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}} \left[\frac{1}{2} \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \hat{Q}(\mathbf{s}_t, \mathbf{a}_t) \right)^2 \right], \tag{7}$$

其中，

$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \, \mathbb{E}_{\mathbf{s}_{t+1} \sim p} \left[V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right], \tag{8}$$

同样可以用随机梯度进行优化：

$$\hat{\nabla}_\theta J_Q(\theta) = \nabla_\theta Q_\theta(\mathbf{a}_t, \mathbf{s}_t) \left(Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - r(\mathbf{s}_t, \mathbf{a}_t) - \gamma V_{\bar{\psi}}(\mathbf{s}_{t+1}) \right). \tag{9}$$

更新使用目标价值网络 $V_{\bar{\psi}}$，其中$\bar{\psi}$ 可以取为价值网络权重的指数移动平均。已有研究表明，这能稳定训练 (Mnih et al., 2015)。另一种做法是定期将目标权重更新为当前价值函数的权重(见附录 E)。最后，策略参数可通过直接最小化式 4 中的期望 KL 散度来学习：

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[\mathrm{D}_{\mathrm{KL}} \left(\pi_\phi(\cdot | \mathbf{s}_t) \,\middle\|\, \frac{\exp\left(Q_\theta(\mathbf{s}_t, \cdot) \right)}{Z_\theta(\mathbf{s}_t)} \right) \right]. \tag{10}$$

最小化 J_π 有多种方式。策略梯度方法的典型做法是使用似然比梯度估计器 (Williams, 1992)，它无需通过策略网络和目标密度网络反向传播梯度。然而，在本文中，目标密度是由神经网络表示的可微 Q 函数，因此采用重参数化技巧更为方便，并能得到方差较低的估计器。为此，我们通过神经网络变换对策略进行重参数化：

$$\mathbf{a}_t = f_\phi(\epsilon_t; \mathbf{s}_t), \tag{11}$$

Algorithm 1 Soft Actor-Critic

Initialize parameter vectors ψ, $\bar{\psi}$, θ, ϕ.
for each iteration **do**
  for each environment step **do**
    $\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t | \mathbf{s}_t)$
    $\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$
  end for
  for each gradient step **do**
    $\psi \leftarrow \psi - \lambda_V \hat{\nabla}_\psi J_V(\psi)$
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i)$ for $i \in \{1, 2\}$
    $\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$
    $\bar{\psi} \leftarrow \tau \psi + (1 - \tau) \bar{\psi}$
  end for
end for

其中，ϵ_t 是从某个固定分布(例如球形高斯分布)采样的输入噪声向量。现在可以将式 10 的目标改写为

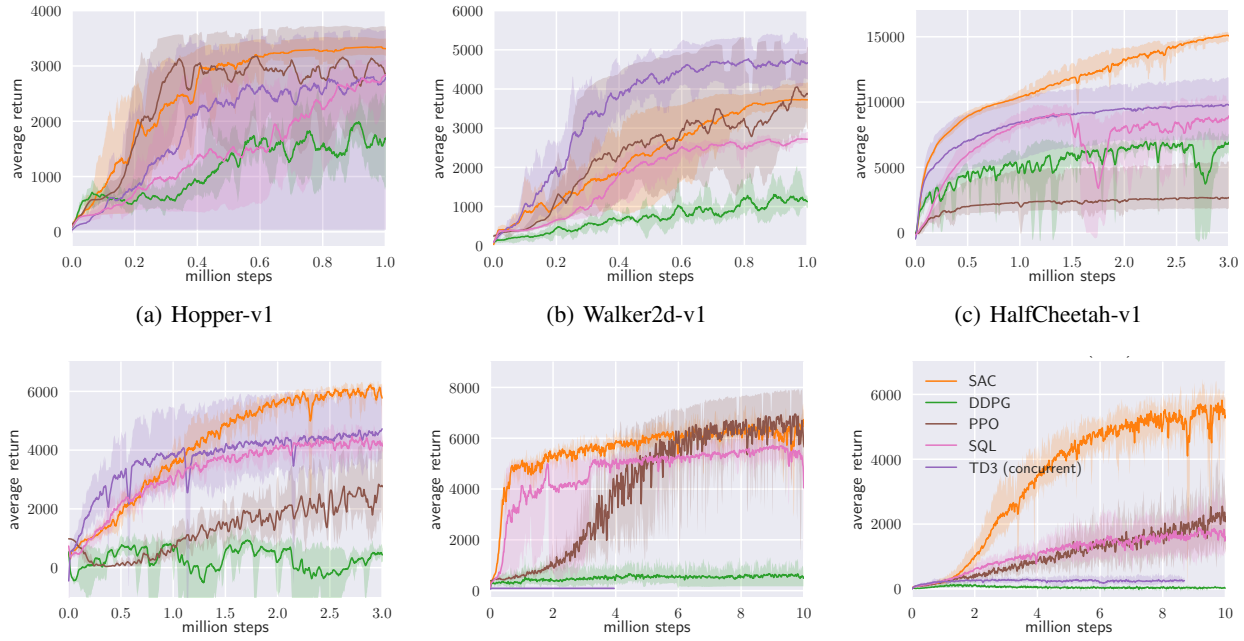
$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}, \epsilon_t \sim \mathcal{N}} \left[\log \pi_\phi(f_\phi(\epsilon_t; \mathbf{s}_t) | \mathbf{s}_t) - Q_\theta(\mathbf{s}_t, f_\phi(\epsilon_t; \mathbf{s}_t)) \right], \tag{12}$$

其中，π_ϕ 由 f_ϕ 隐式定义。由于配分函数与 ϕ 无关，因此可以省略。式 12 的梯度可以近似为

$$\hat{\nabla}_\phi J_\pi(\phi) = \nabla_\phi \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) + \left(\nabla_{\mathbf{a}_t} \log \pi_\phi(\mathbf{a}_t | \mathbf{s}_t) - \nabla_{\mathbf{a}_t} Q(\mathbf{s}_t, \mathbf{a}_t) \right) \nabla_\phi f_\phi(\epsilon_t; \mathbf{s}_t), \tag{13}$$

其中，$\mathbf{a}_t$ 在 $f_\phi(\epsilon_t$ 处计算；$\mathbf{s}_t)$。这一无偏梯度估计器将 DDPG 风格的策略梯度 (Lillicrap et al., 2015) 扩展到了任意便于计算的随机策略。

算法还采用两个 Q 函数，以减轻策略改进步骤中的正偏差；已有研究表明，这种偏差会降低基于价值的方法的性能 (Hasselt, 2010; Fujimoto et al., 2018)。具体而言，我们用参数 θ_i 表示两个 Q 函数，并独立训练它们以优化 $J_Q(\theta_i)$。然后，按照 Fujimoto et al. (2018) 的建议，在式 6 的价值梯度和式 13 的策略梯度中使用两个 Q 函数的较小值。虽然仅用一个 Q 函数也能学习包括 21 维 Humanoid 在内的复杂任务，但我们发现，采用两个 Q 函数能显著加快训练，尤其是在较难的任务上。完整算法见算法 1。该方法交替执行两步：用当前策略与环境交互、收集经验；从经验回放缓冲区采样批次，利用随机梯度更新函数近似器。实践中，每执行一个环境步，随后进行一次或多次梯度更新(全部超参数见附录 D



(a) Hopper-v1 (b) Walker2d-v1 (c) HalfCheetah-v1

(d) Ant-v1 (e) Humanoid-v1 (f) Humanoid (rllab)

Figure 1. Training curves on continuous control benchmarks. Soft actor-critic (yellow) performs consistently across all tasks and outperforming both on-policy and off-policy methods in the most challenging tasks.

for all hyperparameter). Using off-policy data from a replay buffer is feasible because both value estimators and the policy can be trained entirely on off-policy data. The algorithm is agnostic to the parameterization of the policy, as long as it can be evaluated for any arbitrary state-action tuple.

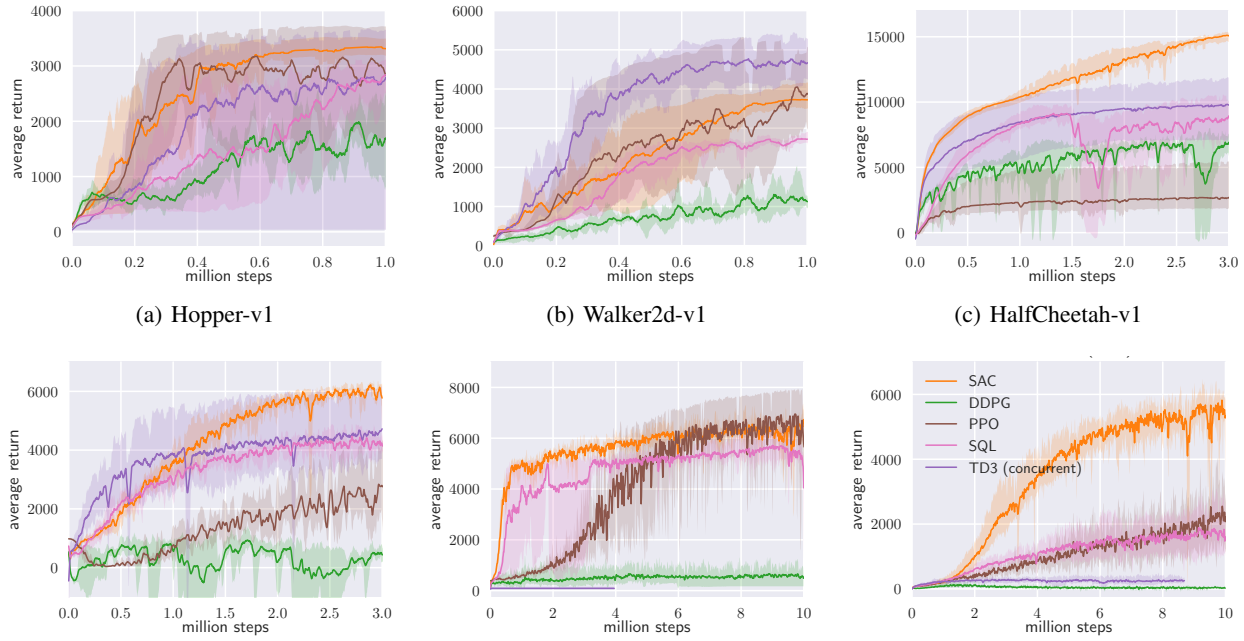
5. Experiments

The goal of our experimental evaluation is to understand how the sample complexity and stability of our method compares with prior off-policy and on-policy deep reinforcement learning algorithms. We compare our method to prior techniques on a range of challenging continuous control tasks from the OpenAI gym benchmark suite (Brockman et al., 2016) and also on the rllab implementation of the Humanoid task (Duan et al., 2016). Although the easier tasks can be solved by a wide range of different algorithms, the more complex benchmarks, such as the 21-dimensional Humanoid (rllab), are exceptionally difficult to solve with off-policy algorithms (Duan et al., 2016). The stability of the algorithm also plays a large role in performance: easier tasks make it more practical to tune hyperparameters to achieve good results, while the already narrow basins of effective hyperparameters become prohibitively small for the more sensitive algorithms on the hardest benchmarks, leading to poor performance (Gu et al., 2016).

We compare our method to deep deterministic policy gradient (DDPG) (Lillicrap et al., 2015), an algorithm that is regarded as one of the more efficient off-policy deep RL methods (Duan et al., 2016); proximal policy optimization (PPO) (Schulman et al., 2017b), a stable and effective on-policy policy gradient algorithm; and soft Q-learning (SQL) (Haarnoja et al., 2017), a recent off-policy algorithm for learning maximum entropy policies. Our SQL implementation also includes two Q-functions, which we found to improve its performance in most environments. We additionally compare to twin delayed deep deterministic policy gradient algorithm (TD3) (Fujimoto et al., 2018), using the author-provided implementation. This is an extension to DDPG, proposed concurrently to our method, that first applied the double Q-learning trick to continuous control along with other improvements. We have included trust region path consistency learning (Trust-PCL) (Nachum et al., 2017b) and two other variants of SAC in Appendix E. We turned off the exploration noise for evaluation for DDPG and PPO. For maximum entropy algorithms, which do not explicitly inject exploration noise, we either evaluated with the exploration noise (SQL) or use the mean action (SAC). The source code of our SAC implementation[1] and videos[2] are available online.

[1] github.com/haarnoja/sac

[2] sites.google.com/view/soft-actor-critic



(a) Hopper-v1 (b) Walker2d-v1 (c) HalfCheetah-v1

(d) Ant-v1 (e) Humanoid-v1 (f) Humanoid (rllab)

图 1. 连续控制基准上的训练曲线。SAC(黄色)在所有任务上都表现稳定，并在最具挑战性的任务上优于同策略和离策略方法。

)。使用经验回放缓冲区中的离策略数据是可行的，因为价值估计器与策略都可以完全采用离策略数据进行训练。只要能对任意状态—动作元组计算策略，该算法就不依赖于具体的策略参数化形式。

5. 实验

实验旨在比较所提方法与以往同策略、离策略深度强化学习算法的样本复杂度及稳定性。我们在 OpenAI Gym 基准中的多种复杂连续控制任务 (Brockman et al., 2016)，以及 rllab 实现的 Humanoid 任务 (Duan et al., 2016) 上进行比较。多种算法都能解决较容易的任务，但对 21 维 Humanoid(rllab)等复杂基准而言，离策略算法尤其难以取得良好结果 (Duan et al., 2016)。算法稳定性也显著影响性能：在简单任务中，通过调节超参数获得良好结果较为可行；而在最难的基准上，敏感算法原本就很窄的有效超参数区域会变得极其狭小，从而导致较差性能 (Gu et al., 2016)。

我们将所提方法与深度确定性策略梯度 (DDPG)(Lillicrap et al., 2015) 比较，该算法被视为效率较高的离策略深度强化学习方法之一 (Duan et al., 2016)；也与稳定有效的同策略策略梯度算法 PPO (Schulman et al., 2017b)，以及用于学习最大熵策略的近期离策略算法软 Q 学习(SQL)(Haarnoja et al., 2017) 比较。我们的 SQL 实现同样包含两个 Q 函数，这能改善其在多数环境中的表现。此外，我们采用作者提供的实现，与同期提出的 TD3 算法 (Fujimoto et al., 2018) 比较。该方法扩展了 DDPG，首次将双 Q 学习技巧用于连续控制，并作出了其他改进。附录 E 还包含 Trust-PCL (Nachum et al., 2017b) 及另外两种 SAC 变体。评估 DDPG 和 PPO 时，我们关闭探索噪声。对于不显式注入探索噪声的最大熵算法，我们在评估 SQL 时保留探索噪声，在评估 SAC 时采用均值动作。SAC 实现的源代码[1]和视频[2]均已在线公开。

[1] github.com/haarnoja/sac
[2] sites.google.com/view/soft-actor-critic

5.1. Comparative Evaluation

Figure 1 shows the total average return of evaluation rollouts during training for DDPG, PPO, and TD3. We train five different instances of each algorithm with different random seeds, with each performing one evaluation rollout every 1000 environment steps. The solid curves corresponds to the mean and the shaded region to the minimum and maximum returns over the five trials.

The results show that, overall, SAC performs comparably to the baseline methods on the easier tasks and outperforms them on the harder tasks with a large margin, both in terms of learning speed and the final performance. For example, DDPG fails to make any progress on Ant-v1, Humanoid-v1, and Humanoid (rllab), a result that is corroborated by prior work (Gu et al., 2016; Duan et al., 2016). SAC also learns considerably faster than PPO as a consequence of the large batch sizes PPO needs to learn stably on more high-dimensional and complex tasks. Another maximum entropy RL algorithm, SQL, can also learn all tasks, but it is slower than SAC and has worse asymptotic performance. The quantitative results attained by SAC in our experiments also compare very favorably to results reported by other methods in prior work (Duan et al., 2016; Gu et al., 2016; Henderson et al., 2017), indicating that both the sample efficiency and final performance of SAC on these benchmark tasks exceeds the state of the art. All hyperparameters used in this experiment for SAC are listed in Appendix D.

5.2. Ablation Study

The results in the previous section suggest that algorithms based on the maximum entropy principle can outperform conventional RL methods on challenging tasks such as the humanoid tasks. In this section, we further examine which particular components of SAC are important for good performance. We also examine how sensitive SAC is to some of the most important hyperparameters, namely reward scaling and target value update smoothing constant.

Stochastic vs. deterministic policy. Soft actor-critic learns stochastic policies via a maximum entropy objective. The entropy appears in both the policy and value function. In the policy, it prevents premature convergence of the policy variance (Equation 10). In the value function, it encourages exploration by increasing the value of regions of state space that lead to high-entropy behavior (Equation 5). To compare how the stochasticity of the policy and entropy maximization affects the performance, we compare to a deterministic variant of SAC that does not maximize the entropy and that closely resembles DDPG, with the exception of having two Q-functions, using hard target updates, not having a separate target actor, and using fixed rather than learned exploration noise. Figure 2 compares five individual runs with both variants, initialized with different random seeds. Soft actor-critic performs much more consistently, while the deterministic variant exhibits very high variability across seeds, indicating substantially worse stability. As evident from the figure, learning a stochastic policy with entropy maximization can drastically stabilize training. This becomes especially important with harder tasks, where tuning hyperparameters is challenging. In this comparison, we updated the target value network weights with hard updates, by periodically overwriting the target network parameters to match the current value network (see Appendix E for a comparison of average performance on all benchmark tasks).

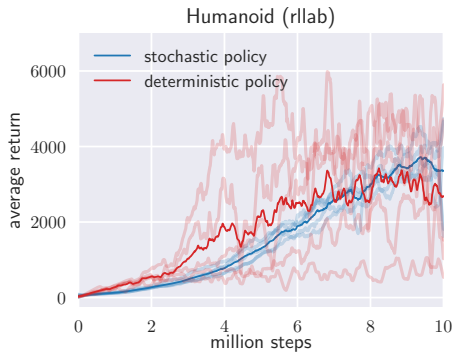


Figure 2. Comparison of SAC (blue) and a deterministic variant of SAC (red) in terms of the stability of individual random seeds on the Humanoid (rllab) benchmark. The comparison indicates that stochasticity can stabilize training as the variability between the seeds becomes much higher with a deterministic policy.

Policy evaluation. Since SAC converges to stochastic policies, it is often beneficial to make the final policy deterministic at the end for best performance. For evaluation, we approximate the maximum a posteriori action by choosing the mean of the policy distribution. Figure 3(a) compares training returns to evaluation returns obtained with this strategy indicating that deterministic evaluation can yield better performance. It should be noted that all of the training curves depict the sum of rewards, which is different from the objective optimized by SAC and other maximum entropy RL algorithms, including SQL and Trust-PCL, which maximize also the entropy of the policy.

Reward scale. Soft actor-critic is particularly sensitive to the scaling of the reward signal, because it serves the role of the temperature of the energy-based optimal policy and thus controls its stochasticity. Larger reward magnitudes correspond to lower entries. Figure 3(b) shows how learning performance changes when the reward scale is varied: For small reward magnitudes, the policy becomes nearly uniform, and consequently fails to exploit the reward signal, resulting in substantial degradation of performance. For large reward magnitudes, the model learns quickly at first,

5.1. 对比评估

图 1 展示了 DDPG、PPO 和 TD3 在训练期间评估轨迹的总平均回报。对每种算法，我们采用不同随机种子训练五个实例，每隔 1000 个环境步执行一次评估。实线表示五次试验的均值，阴影表示最小回报与最大回报之间的范围。

总体而言，SAC 在简单任务上的表现与基线相当，而在较难任务上的学习速度和最终性能均大幅超过基线。例如，DDPG 在 Ant-v1、Humanoid-v1 和 Humanoid(rllab)上均未取得进展，已有工作也报告了相同现象 (Gu et al., 2016; Duan et al., 2016)。SAC 的学习速度也明显快于 PPO，因为 PPO 在高维复杂任务上需要很大的批次才能稳定学习。另一种最大熵强化学习算法 SQL 也能学会所有任务，但速度更慢，渐近性能也较差。SAC 的定量结果与已有工作报告的其他方法相比，同样具有明显优势 (Duan et al., 2016; Gu et al., 2016; Henderson et al., 2017)，表明其在这些基准上的样本效率与最终性能超过了当时的先进水平。实验中 SAC 使用的全部超参数见附录 D。

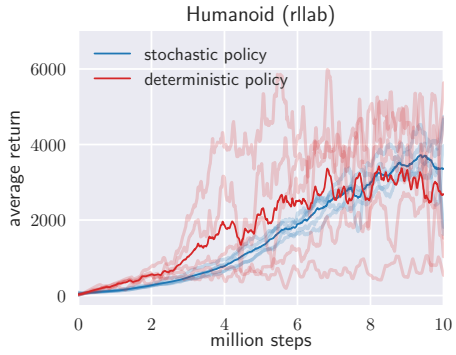


图 2. 在 Humanoid(rllab)基准上，比较 SAC(蓝色)及其确定性变体(红色)在各随机种子下的稳定性。确定性策略在不同种子间的差异明显增大，说明随机性有助于稳定训练。

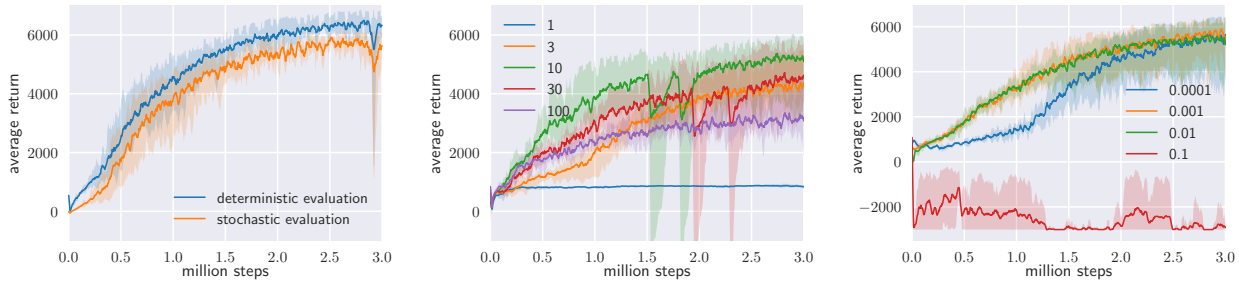
5.2. 消融研究

上一节结果表明，基于最大熵原理的算法能够在 Humanoid 等复杂任务上超过传统强化学习方法。本节进一步分析 SAC 的哪些组成部分对于良好性能至关重要，并检验 SAC 对两个关键超参数的敏感性：奖励缩放和目标价值更新的平滑常数。

随机策略与确定性策略。SAC 通过最大熵目标学习随机策略，熵同时出现在策略函数和价值函数中。在策略中，熵可以防止策略方差过早收敛(式 10)；在价值函数中，它通过提高通向高熵行为的状态区域的价值来促进探索(式 5)。为比较策略随机性和熵最大化对性能的影响，我们构造了一种不最大化熵的确定性 SAC 变体。它与 DDPG 很接近，但采用两个 Q 函数、硬目标更新、没有独立的目标 Actor，并使用固定的探索噪声，而非学习噪声。图 2 比较了两个变体各五次独立运行的结果，运行采用不同的随机种子进行初始化。SAC 的表现更加一致，而确定性变体在不同种子间的差异很大，表明稳定性明显较差。从图中可以看出，通过熵最大化学习随机策略，能够显著稳定训练。这对于超参数难以调节的复杂任务尤其重要。在该对比中，我们采用硬更新：定期将当前价值网络的权重复制到目标价值网络(所有基准上的平均性能对比见附录 E)。

策略评估。由于 SAC 收敛到随机策略，为获得最佳性能，最终评估时采用确定性策略通常更有利。我们选择策略分布的均值来近似最大后验动作。图 3(a) 比较了训练回报与按此方式得到的评估回报，表明确定性评估能够取得更好性能。所有训练曲线展示的都是奖励之和；它与 SAC、SQL、Trust-PCL 等最大熵强化学习算法实际优化的目标不同，后者还要最大化策略熵。

奖励尺度。SAC 对奖励信号的缩放尤其敏感，因为它起到基于能量的最优策略温度参数的作用，从而控制策略随机性。较大的奖励幅度对应较低的熵。图 3(b) 展示了改变奖励尺度时的学习表现：奖励幅度较小时，策略接近均匀分布，无法充分利用奖励信号，因而性能明显下降；奖励幅度较大时，模型起初学习较快，



(a) Evaluation (b) Reward Scale (c) Target Smoothing Coefficient (τ)

Figure 3. Sensitivity of soft actor-critic to selected hyperparameters on Ant-v1 task. (a) Evaluating the policy using the mean action generally results in a higher return. Note that the policy is trained to maximize also the entropy, and the mean action does not, in general, correspond the optimal action for the maximum return objective. (b) Soft actor-critic is sensitive to reward scaling since it is related to the temperature of the optimal policy. The optimal reward scale varies between environments, and should be tuned for each task separately. (c) Target value smoothing coefficient τ is used to stabilize training. Fast moving target (large τ) can result in instabilities (red), whereas slow moving target (small τ) makes training slower (blue).

but the policy then becomes nearly deterministic, leading to poor local minima due to lack of adequate exploration. With the right reward scaling, the model balances exploration and exploitation, leading to faster learning and better asymptotic performance. In practice, we found reward scale to be the only hyperparameter that requires tuning, and its natural interpretation as the inverse of the temperature in the maximum entropy framework provides good intuition for how to adjust this parameter.

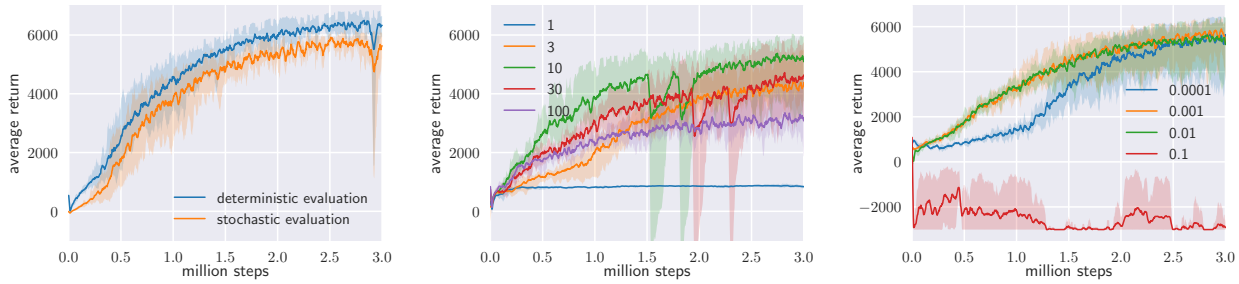
Target network update. It is common to use a separate target value network that slowly tracks the actual value function to improve stability. We use an exponentially moving average, with a smoothing constant τ, to update the target value network weights as common in the prior work (Lillicrap et al., 2015; Mnih et al., 2015). A value of one corresponds to a hard update where the weights are copied directly at every iteration and zero to not updating the target at all. In Figure 3(c), we compare the performance of SAC when τ varies. Large τ can lead to instabilities while small τ can make training slower. However, we found the range of suitable values of τ to be relatively wide and we used the same value (0.005) across all of the tasks. In Figure 4 (Appendix E) we also compare to another variant of SAC, where instead of using exponentially moving average, we copy over the current network weights directly into the target network every 1000 gradient steps. We found this variant to benefit from taking more than one gradient step between the environment steps, which can improve performance but also increases the computational cost.

6. Conclusion

We present soft actor-critic (SAC), an off-policy maximum entropy deep reinforcement learning algorithm that provides sample-efficient learning while retaining the benefits of entropy maximization and stability. Our theoretical results derive soft policy iteration, which we show to converge to the optimal policy. From this result, we can formulate a soft actor-critic algorithm, and we empirically show that it outperforms state-of-the-art model-free deep RL methods, including the off-policy DDPG algorithm and the on-policy PPO algorithm. In fact, the sample efficiency of this approach actually exceeds that of DDPG by a substantial margin. Our results suggest that stochastic, entropy maximizing reinforcement learning algorithms can provide a promising avenue for improved robustness and stability, and further exploration of maximum entropy methods, including methods that incorporate second order information (e.g., trust regions (Schulman et al., 2015)) or more expressive policy classes is an exciting avenue for future work.

Acknowledgments

We would like to thank Vitchyr Pong for insightful discussions and help in implementing our algorithm as well as providing the DDPG baseline code; Ofir Nachum for offering support in running Trust-PCL experiments; and George Tucker for his valuable feedback on an early version of this paper. This work was supported by Siemens and Berkeley DeepDrive.



(a) Evaluation (b) Reward Scale (c) Target Smoothing Coefficient (τ)

图 3. SAC 在 Ant-v1 任务上对若干超参数的敏感性。(a) 采用均值动作评估，通常能获得更高回报。训练时策略还会最大化熵，而均值动作一般并不对应最大回报目标下的最优动作。(b) 奖励尺度与最优策略的温度有关，因此 SAC 对奖励缩放敏感。不同环境的最优奖励尺度不同，需要分别调节。(c) 目标价值平滑系数 τ 用于稳定训练。目标变化过快(τ 较大)可能导致不稳定(红色)；目标变化较慢(τ 较小)则会减缓学习(蓝色)。

但随后策略会变得近乎确定，并因探索不足而陷入较差的局部极小值。合适的奖励缩放能平衡探索与利用，从而加快学习并改善渐近性能。实践中，我们发现奖励尺度是唯一需要调节的超参数；将其自然地解释为最大熵框架中温度的倒数，有助于直观判断该如何调整。

目标网络更新。常见做法是设置独立目标价值网络，使其缓慢跟踪实际价值函数，以提高稳定性。与已有工作相同，我们采用平滑常数为 τ 的指数移动平均更新目标价值网络权重 (Lillicrap et al., 2015; Mnih et al., 2015)。取一时，相当于每轮直接复制权重的硬更新；取零时，目标网络完全不更新。图 3(c) 比较了改变 τ 时 SAC 的表现：τ 较大可能导致不稳定，τ 较小则可能减缓训练。不过，τ 的合适取值范围相对较宽，我们在所有任务上均使用同一取值(0.005)。附录 E 的图 4 还比较了另一种 SAC 变体：不用指数移动平均，而是每隔 1000 次梯度更新，将当前网络权重直接复制到目标网络。我们发现，在环境步之间执行多次梯度更新有利于这一变体，可以提高性能，但也会增加计算成本。

6. 结论

本文提出软 Actor-Critic(SAC)：一种离策略最大熵深度强化学习算法，兼具较高样本效率、熵最大化的优势与训练稳定性。理论上，我们推导了软策略迭代，并证明其收敛到最优策略。在此基础上构造 SAC 算法，实验表明，它优于当时先进的无模型深度强化学习方法，包括离策略 DDPG 和同策略 PPO。其样本效率也大幅超过 DDPG。结果表明，采用随机策略并最大化熵的强化学习算法，是提高鲁棒性和稳定性的有前景方向。后续值得进一步研究最大熵方法，包括引入二阶信息，例如信赖域 (Schulman et al., 2015)，或采用表达能力更强的策略类。

致谢

感谢 Vitchyr Pong 提供富有启发的讨论、协助实现算法，并提供 DDPG 基线代码；感谢 Ofir Nachum 协助运行 Trust-PCL 实验；感谢 George Tucker 对论文早期版本提出宝贵意见。本研究得到 Siemens 和 Berkeley DeepDrive 的支持。

References

Barto, A. G., Sutton, R. S., and Anderson, C. W. Neuronlike adaptive elements that can solve difficult learning control problems. *IEEE transactions on systems, man, and cybernetics*, pp. 834–846, 1983.

Bhatnagar, S., Precup, D., Silver, D., Sutton, R. S., Maei, H. R., and Szepesvári, C. Convergent temporal-difference learning with arbitrary smooth function approximation. In *Advances in Neural Information Processing Systems (NIPS)*, pp. 1204–1212, 2009.

Brockman, G., Cheung, V., Pettersson, L., Schneider, J., Schulman, J., Tang, J., and Zaremba, W. OpenAI gym. *arXiv preprint arXiv:1606.01540*, 2016.

Duan, Y., Chen, X. Houthooft, R., Schulman, J., and Abbeel, P. Benchmarking deep reinforcement learning for continuous control. In *International Conference on Machine Learning (ICML)*, 2016.

Fox, R., Pakman, A., and Tishby, N. Taming the noise in reinforcement learning via soft updates. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2016.

Fujimoto, S., van Hoof, H., and Meger, D. Addressing function approximation error in actor-critic methods. *arXiv preprint arXiv:1802.09477*, 2018.

Gruslys, A., Azar, M. G., Bellemare, M. G., and Munos, R. The reactor: A sample-efficient actor-critic architecture. *arXiv preprint arXiv:1704.04651*, 2017.

Gu, S., Lillicrap, T., Ghahramani, Z., Turner, R. E., and Levine, S. Q-prop: Sample-efficient policy gradient with an off-policy critic. *arXiv preprint arXiv:1611.02247*, 2016.

Haarnoja, T., Tang, H., Abbeel, P., and Levine, S. Reinforcement learning with deep energy-based policies. In *International Conference on Machine Learning (ICML)*, pp. 1352–1361, 2017.

Hasselt, H. V. Double Q-learning. In *Advances in Neural Information Processing Systems (NIPS)*, pp. 2613–2621, 2010.

Heess, N., Wayne, G., Silver, D., Lillicrap, T., Erez, T., and Tassa, Y. Learning continuous control policies by stochastic value gradients. In *Advances in Neural Information Processing Systems (NIPS)*, pp. 2944–2952, 2015.

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., and Meger, D. Deep reinforcement learning that matters. *arXiv preprint arXiv:1709.06560*, 2017.

Kingma, D. and Ba, J. Adam: A method for stochastic optimization. In *International Conference for Learning Presentations (ICLR)*, 2015.

Levine, S. and Koltun, V. Guided policy search. In *International Conference on Machine Learning (ICML)*, pp. 1–9, 2013.

Levine, S., Finn, C., Darrell, T., and Abbeel, P. End-to-end training of deep visuomotor policies. *Journal of Machine Learning Research*, 17(39):1–40, 2016.

Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015.

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., and Riedmiller, M. Playing atari with deep reinforcement learning. *arXiv preprint arXiv:1312.5602*, 2013.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540): 529–533, 2015.

Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T. P., Harley, T., Silver, D., and Kavukcuoglu, K. Asynchronous methods for deep reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2016.

Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. Bridging the gap between value and policy based reinforcement learning. In *Advances in Neural Information Processing Systems (NIPS)*, pp. 2772–2782, 2017a.

Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. Trust-PCL: An off-policy trust region method for continuous control. *arXiv preprint arXiv:1707.01891*, 2017b.

O'Donoghue, B., Munos, R., Kavukcuoglu, K., and Mnih, V. PGQ: Combining policy gradient and Q-learning. *arXiv preprint arXiv:1611.01626*, 2016.

Peters, J. and Schaal, S. Reinforcement learning of motor skills with policy gradients. *Neural networks*, 21(4):682–697, 2008.

Rawlik, K., Toussaint, M., and Vijayakumar, S. On stochastic optimal control and reinforcement learning by approximate inference. *Robotics: Science and Systems (RSS)*, 2012.

Schulman, J., Levine, S., Abbeel, P., Jordan, M. I., and Moritz, P. Trust region policy optimization. In *International Conference on Machine Learning (ICML)*, pp. 1889–1897, 2015.

参考文献

Barto, A. G., Sutton, R. S., and Anderson, C. W. 能够解决复杂学习控制问题的类神经元自适应单元。IEEE trans actions on systems, man, and cybernetics, pp. 834–846, 19 83.

Bhatnagar, S., Precup, D., Silver, D., Sutton, R. S., Maei, H. R., and Szepesvári, C. 采用任意光滑函数近似的收敛时序差分学习。In Advances in Neural Information Proce ssing Systems (NIPS), pp. 1204–1212, 2009.

Brockman, G., Cheung, V., Pettersson, L., Schneider, J., S chulman, J., Tang, J., and Zaremba, W. OpenAI Gym。ar Xiv preprint arXiv:1606.01540, 2016.

Duan, Y., Chen, X. Houthooft, R., Schulman, J., and Abbe el, P. 连续控制中的深度强化学习基准评测。In Internati onal Conference on Machine Learning (ICML), 2016.

Fox, R., Pakman, A., and Tishby, N. 通过软更新抑制强化学习中的噪声。In Conference on Uncertainty in Artifi cial Intelligence (UAI), 2016.

Fujimoto, S., van Hoof, H., and Meger, D. 处理 Actor-Crit ic 方法中的函数近似误差。arXiv preprint arXiv:1802.09 477, 2018.

Gruslys, A., Azar, M. G., Bellemare, M. G., and Munos, R. Reactor：一种高样本效率的 Actor-Critic 架构。arXiv p reprint arXiv:1704.04651, 2017.

Gu, S., Lillicrap, T., Ghahramani, Z., Turner, R. E., and Le vine, S. Q-prop：采用离策略 Critic 的高样本效率策略梯度。arXiv preprint arXiv:1611.02247, 2016.

Haarnoja, T., Tang, H., Abbeel, P., and Levine, S. 采用深度能量策略的强化学习。In International Conference on Machine Learning (ICML), pp. 1352–1361, 2017.

Hasselt, H. V. 双 Q 学习。In Advances in Neural Informa tion Processing Systems (NIPS), pp. 2613–2621, 2010.

Heess, N., Wayne, G., Silver, D., Lillicrap, T., Erez, T., an d Tassa, Y. 通过随机价值梯度学习连续控制策略。In A dvances in Neural Information Processing Systems (NIPS), pp. 2944–2952, 2015.

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., and Meger, D. 深度强化学习中真正重要的因素。ar Xiv preprint arXiv:1709.06560, 2017.

Kingma, D. and Ba, J. Adam：一种随机优化方法。In In ternational Conference for Learning Presentations (ICLR), 2015.

Levine, S. and Koltun, V. 引导策略搜索。In International Conference on Machine Learning (ICML), pp. 1–9, 2013.

Levine, S., Finn, C., Darrell, T., and Abbeel, P. 深度视觉运动策略的端到端训练。Journal of Machine Learning R esearch, 17(39):1–40, 2016.

Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. 采用深度强化学习的连续控制。arXiv preprint arXiv:1509.02971, 2015.

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antono glou, I., Wierstra, D., and Riedmiller, M. 使用深度强化学习玩 Atari 游戏。arXiv preprint arXiv:1312.5602, 2013.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Venes s, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjela nd, A. K., Ostrovski, G., et al. 通过深度强化学习实现人类水平的控制。Nature, 518(7540): 529–533, 2015.

Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T. P., Harley, T., Silver, D., and Kavukcuoglu, K. 深度强化学习的异步方法。In International Conference on Machin e Learning (ICML), 2016.

Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. 弥合基于价值与基于策略的强化学习之间的差距。In Ad vances in Neural Information Processing Systems (NIPS), pp. 2772–2782, 2017a.

Nachum, O., Norouzi, M., Xu, K., and Schuurmans, D. Tr ust-PCL：用于连续控制的离策略信赖域方法。arXiv pr eprint arXiv:1707.01891, 2017b.

O’Donoghue, B., Munos, R., Kavukcuoglu, K., and Mnih , V. PGQ：结合策略梯度与 Q 学习。arXiv preprint arXi v:1611.01626, 2016.

Peters, J. and Schaal, S. 利用策略梯度对运动技能进行强化学习。Neural networks, 21(4):682– 697, 2008.

Rawlik, K., Toussaint, M., and Vijayakumar, S. 通过近似推断进行随机最优控制与强化学习。Robotics: Science and Systems (RSS), 2012.

Schulman, J., Levine, S., Abbeel, P., Jordan, M. I., and Mo ritz, P. 信赖域策略优化。In International Conference on Machine Learning (ICML), pp. 1889–1897, 2015.

Schulman, J., Abbeel, P., and Chen, X. Equivalence between policy gradients and soft Q-learning. *arXiv preprint arXiv:1704.06440*, 2017a.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017b.

Silver, D., Lever, G., Heess, N., Degris, T., Wierstra, D., and Riedmiller, M. Deterministic policy gradient algorithms. In *International Conference on Machine Learning (ICML)*, 2014.

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., and Hassabis, D. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, Jan 2016. ISSN 0028-0836. Article.

Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.

Thomas, P. Bias in natural actor-critic algorithms. In *International Conference on Machine Learning (ICML)*, pp. 441–448, 2014.

Todorov, E. General duality between optimal control and estimation. In *IEEE Conference on Decision and Control (CDC)*, pp. 4286–4292. IEEE, 2008.

Toussaint, M. Robot trajectory optimization using approximate inference. In *International Conference on Machine Learning (ICML)*, pp. 1049–1056. ACM, 2009.

Williams, R. J. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning*, 8(3-4):229–256, 1992.

Ziebart, B. D. *Modeling purposeful adaptive behavior with the principle of maximum causal entropy*. Carnegie Mellon University, 2010.

Ziebart, B. D., Maas, A. L., Bagnell, J. A., and Dey, A. K. Maximum entropy inverse reinforcement learning. In *AAAI Conference on Artificial Intelligence (AAAI)*, pp. 1433–1438, 2008.

Schulman, J., Abbeel, P., and Chen, X. 策略梯度与软 Q 学习的等价性。arXiv preprint arXiv:1704.06440, 2017a.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. 近端策略优化算法。arXiv preprint arXiv:1707.06347, 2017b.

Silver, D., Lever, G., Heess, N., Degris, T., Wierstra, D., and Riedmiller, M. 确定性策略梯度算法。In International Conference on Machine Learning (ICML), 2014.

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., and Hassabis, D. 通过深度神经网络和树搜索掌握围棋。Nature, 529(7587):484–489, Jan 2016. ISSN 0028-0836. Article.

Sutton, R. S. and Barto, A. G. 强化学习导论，第 1 卷。MIT press Cambridge, 1998.

Thomas, P. 自然 Actor-Critic 算法中的偏差。In International Conference on Machine Learning (ICML), pp. 441–448, 2014.

Todorov, E. 最优控制与估计之间的一般对偶性。In IEEE Conference on Decision and Control (CDC), pp. 4286–4292. IEEE, 2008.

Toussaint, M. 利用近似推断进行机器人轨迹优化。In International Conference on Machine Learning (ICML), pp. 1049–1056. ACM, 2009.

Williams, R. J. 用于联结主义强化学习的简单统计梯度跟随算法。Machine learning, 8(3-4):229–256, 1992.

Ziebart, B. D. 利用最大因果熵原理对有目的的自适应行为建模。Carnegie Mellon University, 2010.

Ziebart, B. D., Maas, A. L., Bagnell, J. A., and Dey, A. K. 最大熵逆强化学习。In AAAI Conference on Artificial Intelligence (AAAI), pp. 1433–1438, 2008.